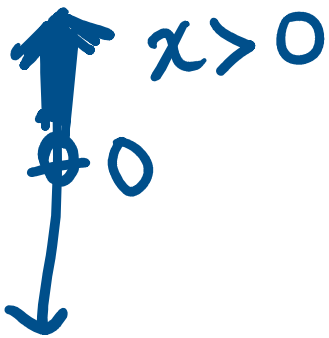
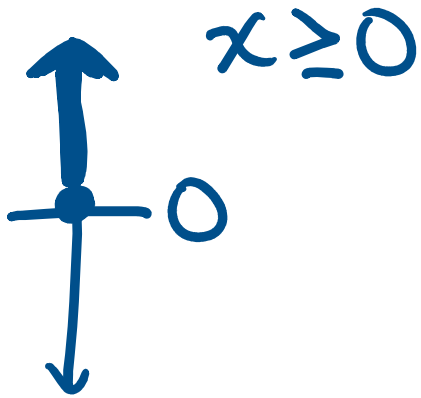
Inequalities are when something is greater than or less than something else. https://simple.wikipedia.org/wiki/Inequality#:

Like equalities, they are true or false.

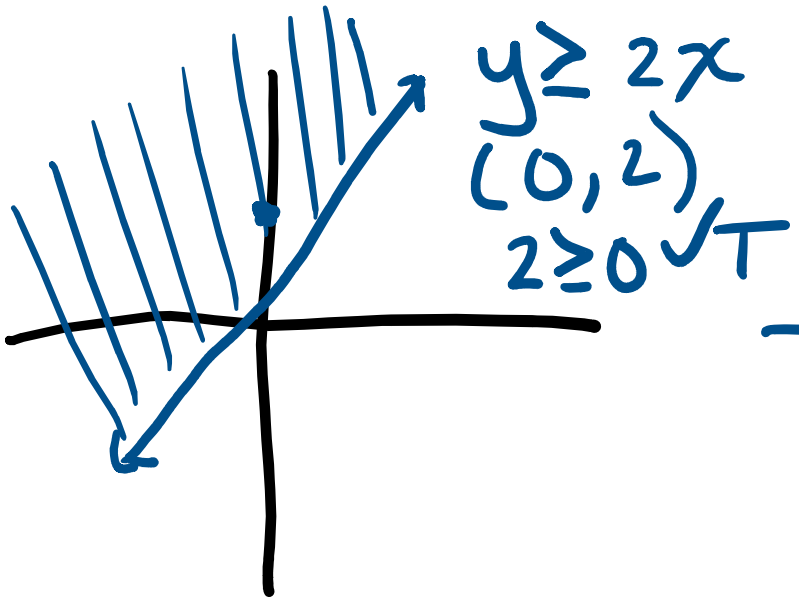
$1 < 2$ ✓T $\qquad$ $2 < 1$ F X

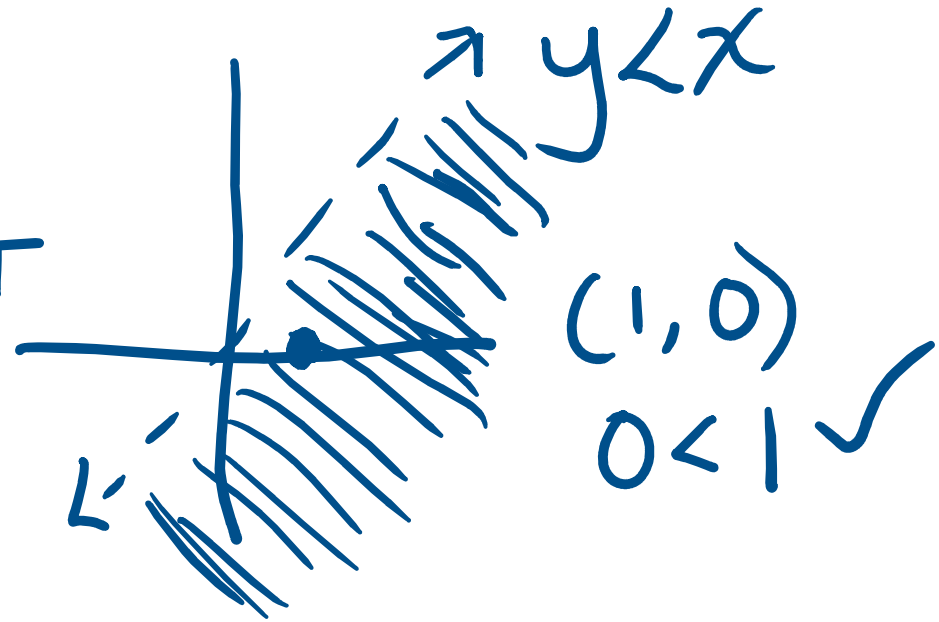
Inequalities can be graphed in one variable.

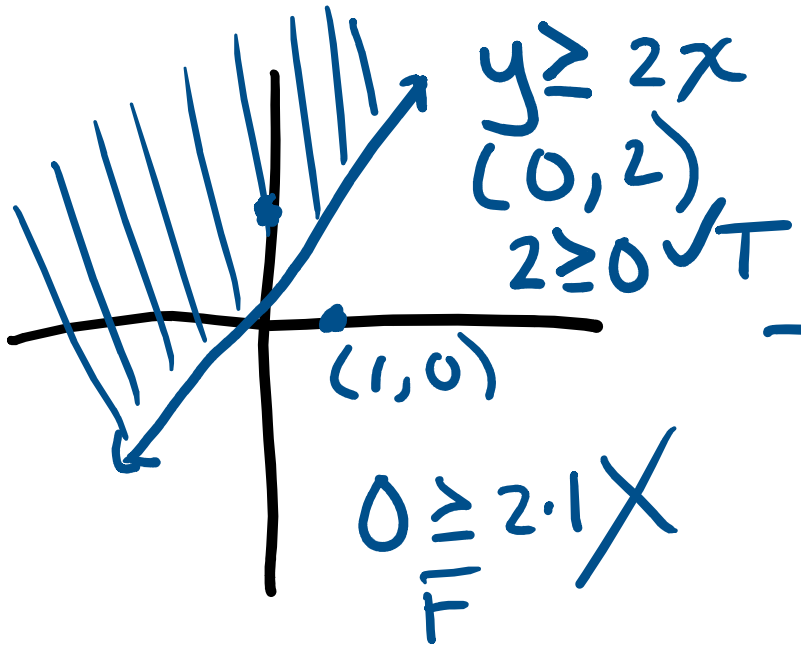


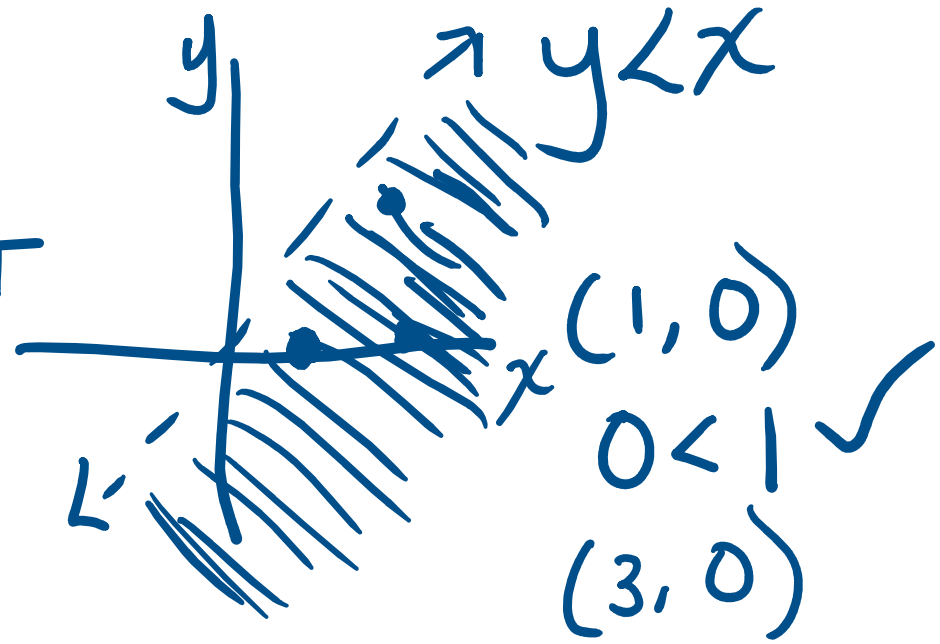


Dashed line is not equal, solid line is equal. Shade the side of the line that makes the inequality true.






y ≥ 2x
(0, 2)
2 ≥ 0 ✓ T
(1, 0)
0 ≥ 2·1 ✗
F


y
y < x
x
(1, 0)
0 < 1 ✓
(3, 0)

If you aren't sure which side of the line to shade, pick an easy point and see if it makes it true. Remember that the order switches as y<x has the y first and the point (x,y) has the x first.

$y > x$

(x, y)

$y > x$

$(0, 1)$

$1 > 0$ ✓

For $y<x$, I think of the output as being less than the input so it would be lower.

Picking a point like (0,2) shows that above the line should not be shaded as 2 is not less than zero. So below the line should be shaded.

For y>x, I think of the output as being greater than the input so it would higher.

Picking a point like (0,2) shows that, indeed, above the line should be shaded.

We can also solve them algebraically. They work just like equalities with 2 exceptions:

When you flip the order, the sign's orientation stays intact. Flipping it like a pancake.

$x<y$ $\quad 1<2 \quad 3>-1 \quad -1>-2 \uparrow$

$y>x$ $\quad 2>1 \quad -1<3 \quad -2<-1 \downarrow$

$x<y$

$y>x$

When you flip the order, the sign's orientation stays intact. You need to preserve the relationship.

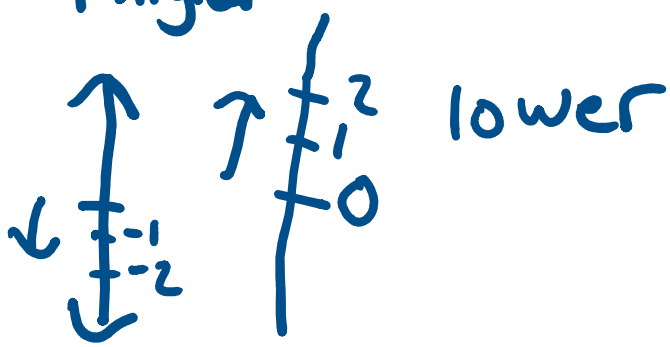
$x<y$ $\quad 1<2 \quad 3>-1 \quad -1>-2 \uparrow$

$y>x$ $\quad 2>1 \quad -1<3 \quad -2<-1 \downarrow$

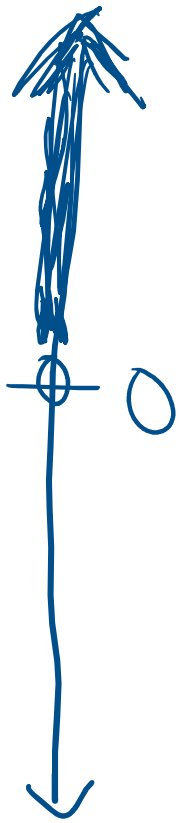
When you multiply by a negative number (multiply by negative one), you switch the orientation of the sign.

$-1(-1>-2)$ $\quad$ -1 higher

$1<2$



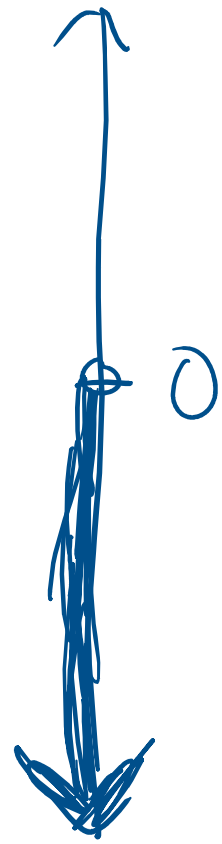
Multiplying by negative one is like changing the direction of the number line.



$x > 0$

$$\boxed{\begin{aligned} x - x &> 0 - x \\ 0 &> -x \end{aligned}}$$

$-x < 0$



For a line:

$$y \lessgtr ax+b$$

$$2x+2y>1$$

Solving inequalities algebraically:

$2x+1>5$

$9<2x+1$

$4-x>1$

$5<3-x$